

- How is lexical information (word position, character, etc.) represented in a recurrent network that learns to produce words?
- How does the network's representation of information change with vocabulary complexity?
- How do the network's weights differ as a function of the complexity of the learned vocabulary (function $\rightarrow$ structure)?

Vocabulary B : DFA=8

hat hate late

hatlatehatehatlatehatehatlatehatehatlatehatehathate

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graph LR; start((ε)) -- h --> h((h)); h -- a --> ha((ha)); ha -- t --> hat((hat)); hat -- e --> hate((hate)); hate -- e --> late((late)); late -- t --> lat((lat)); lat -- a --> la((la)); la -- a --> l((l)); l -- e --> late; late -- e --> hate; style start fill:none,stroke:none; style hate fill:#fff,stroke:#000,stroke-width:2px; style late fill:#fff,stroke:#000,stroke-width:2px;
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Character-level Elman RNN. Input x_t predicts the next character via $h_t = \tanh(W_{zh}x_t + W_{hh}h_{t-1})$; cross-entropy loss.

Loss and OOV collapse together; rollouts become legal vocabulary words. Demo lexicon listed above the panels.

Softmax $P(\text{next char} \mid \text{input so far})$. Red dots = actual next character.

Hidden units over timesteps (input letters; units clustered)

Activations clustered by in-word prefix. Tick color = DFA state.

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Key references: Saffran, Aslin & Newport (1996); Elman (1990); Aslin, Saffran & Newport (1998); Goldwater et al. (2009); Perruchet & Vinter (1998)

Prefix $\times$ prefix correlation. Blocks merge prefixes that share a future.

PCA of **h** by feature; pairwise within/between/shuffle at end of row. DFA $\eta^2 \approx 0.88$; word identity ≈ 0.06 .

One top unit per feature on a shared corpus window.

Mean $\pm$ std, 6 seeds. DFA/character saturate in a few PCs; word identity needs many more. Right: unit SI density (lines).

DFA-labeled prefixes with closed-loop trajectories overlaid. Columns: 2–5 word mixed vocabs (DFA ≤ 12 where available); rows: two seeds.

50 mixed vocabs ($H=100$). Top: CE and word-error curves by DFA size. Bottom: iters to 3% WE and closed-loop PC spectra

10 mixed vocabs spanning DFA size. Low-DFA: coarse bands; larger DFAs: denser rasters

Two DFA bins. Left: final PCA/heuron readout. Right: readout over learning (3/15 PCs). Word identity needs ~ 15 PCs.

Within/between cosine similarity, plus per-unit SI vs DFA size.

Init + DFA 8/9/13 matrices (shared symlog w scale); bottom: W_{gh} / W_{hh} densities (color = DFA size)

Norms, structure, pairwise and 3-node motifs. Larger DFAs shift feedforward.